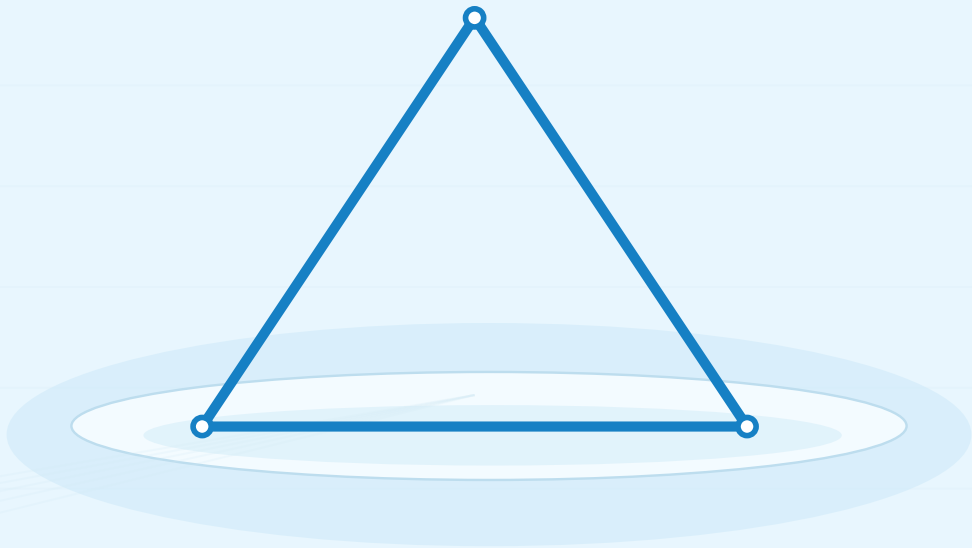
LET'S EXPLORE TRIANGLES

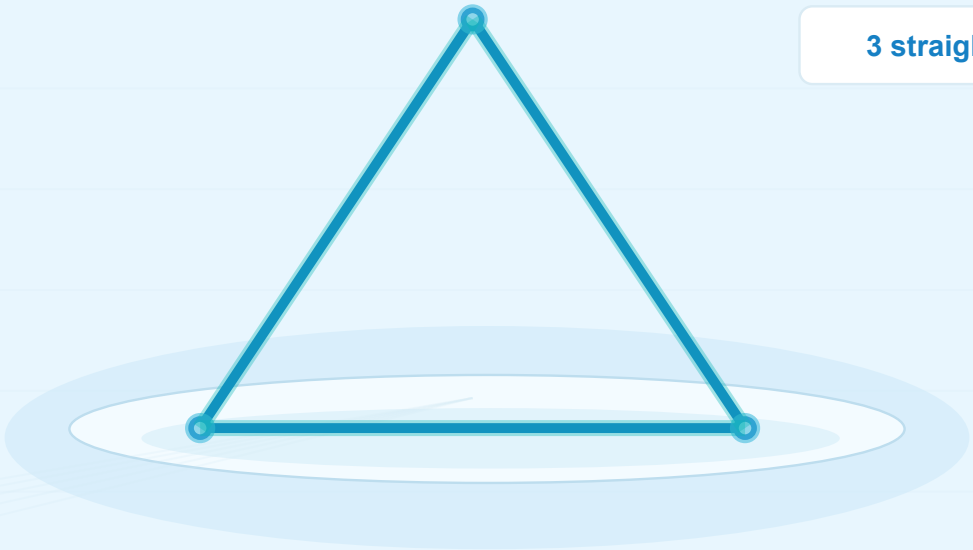
Before we return to the tunnel, learn how to recognise them.



BEGIN LEARNING

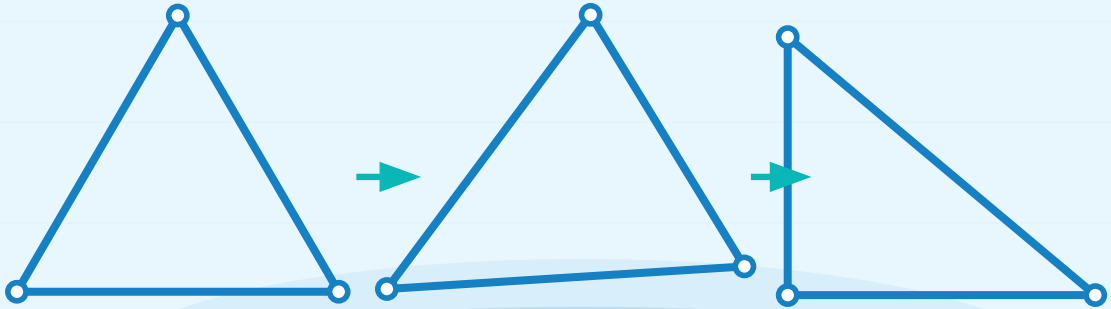


This is a triangle.



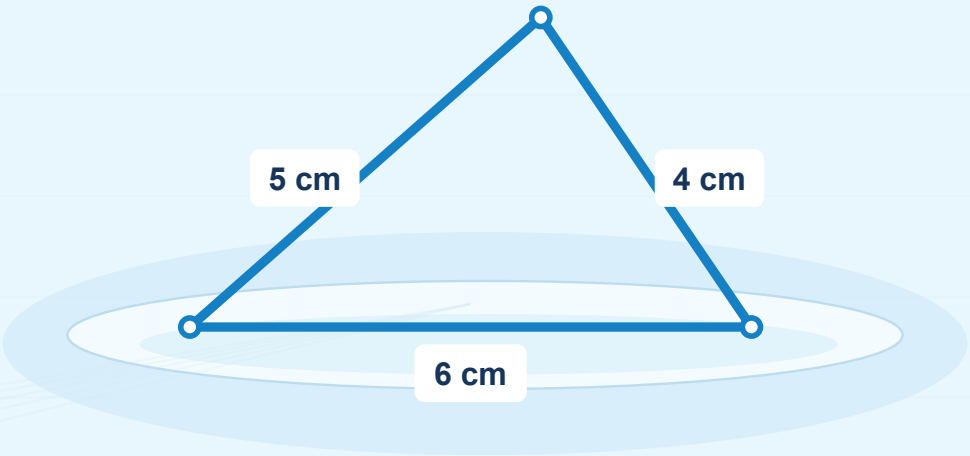
3 straight sides

A closed shape with 3 straight sides is called a triangle.

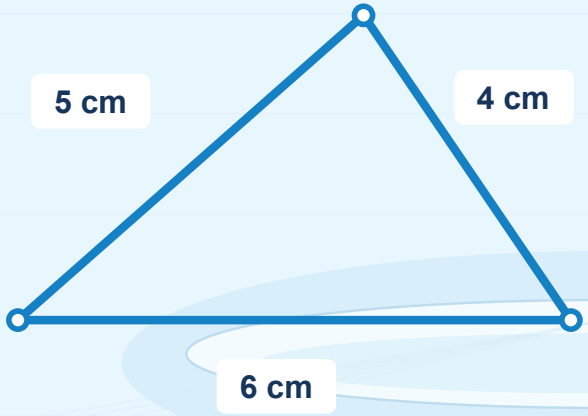


Triangles can be of different types. Let's explore them!

MEASURE MODE

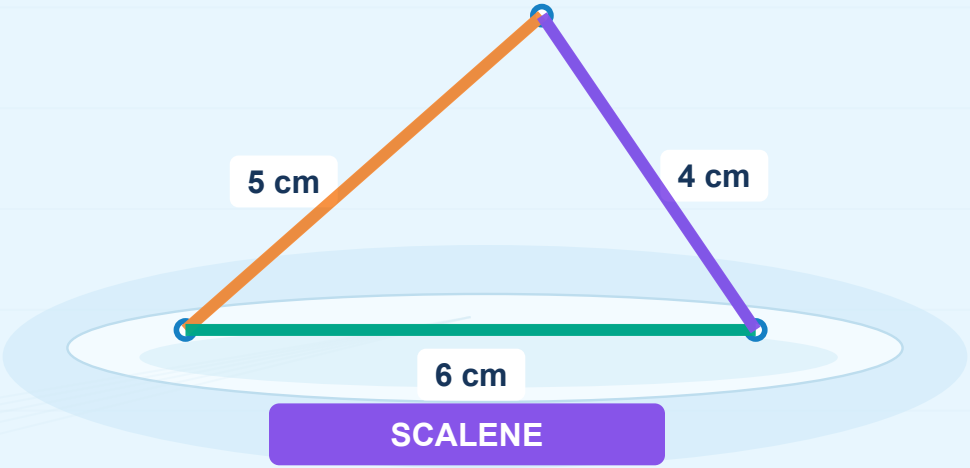


Let's measure the sides of this triangle.

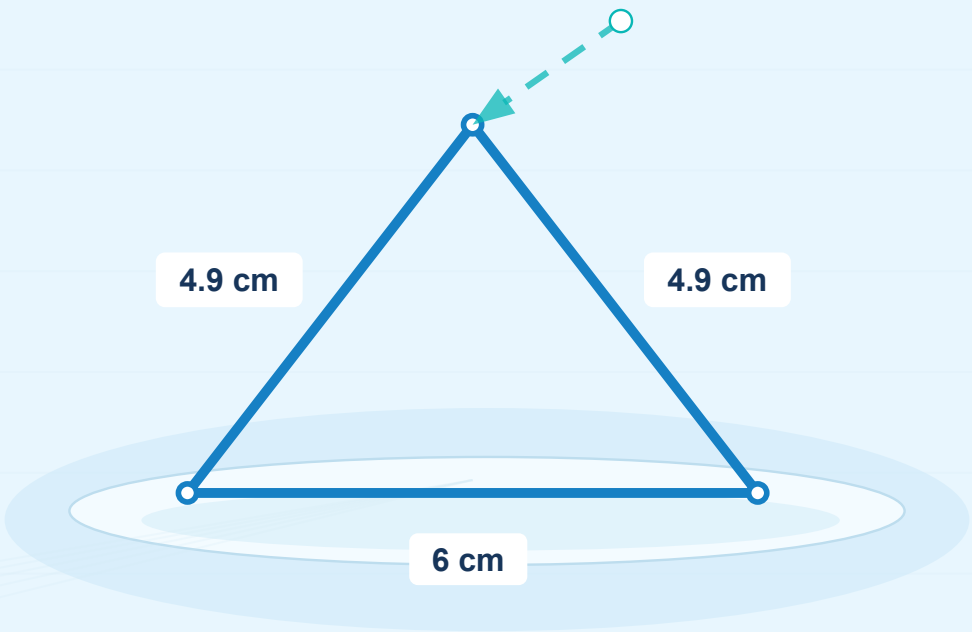


What do you notice about the sides?

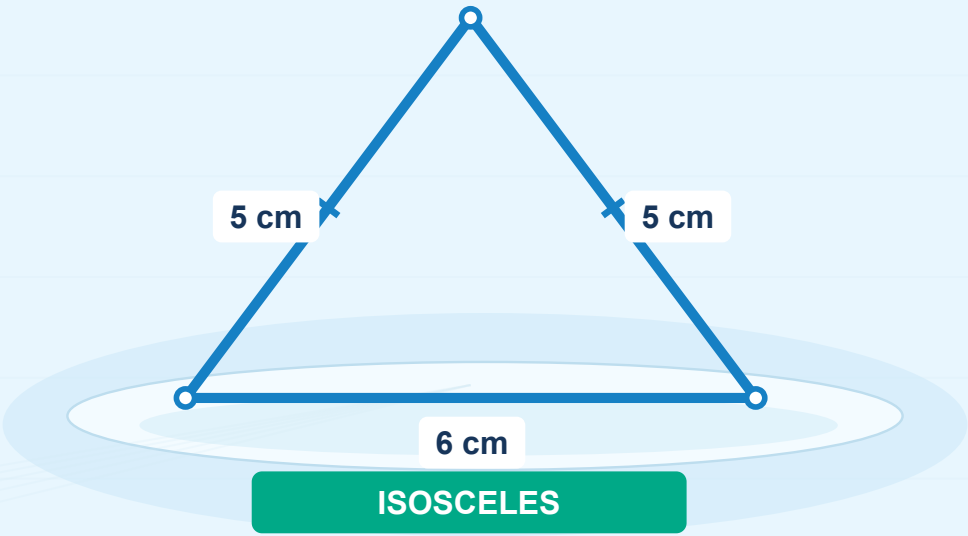
- [x] All three are different
- [ ] Two sides are equal
- [ ] All three are equal



A triangle with all three sides different is called a scalene triangle.



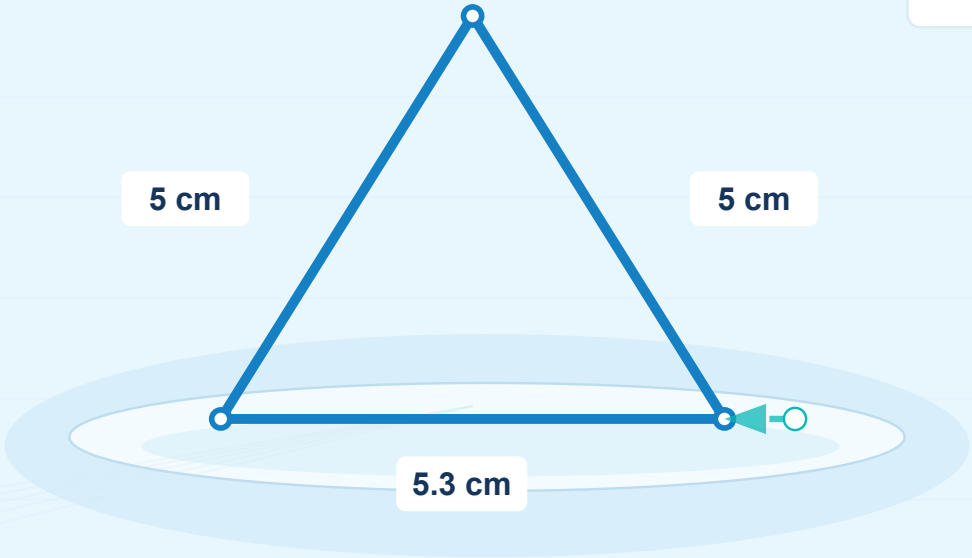
What if two sides were equal?
Drag any vertex to make two sides equal.



ISOSCELES

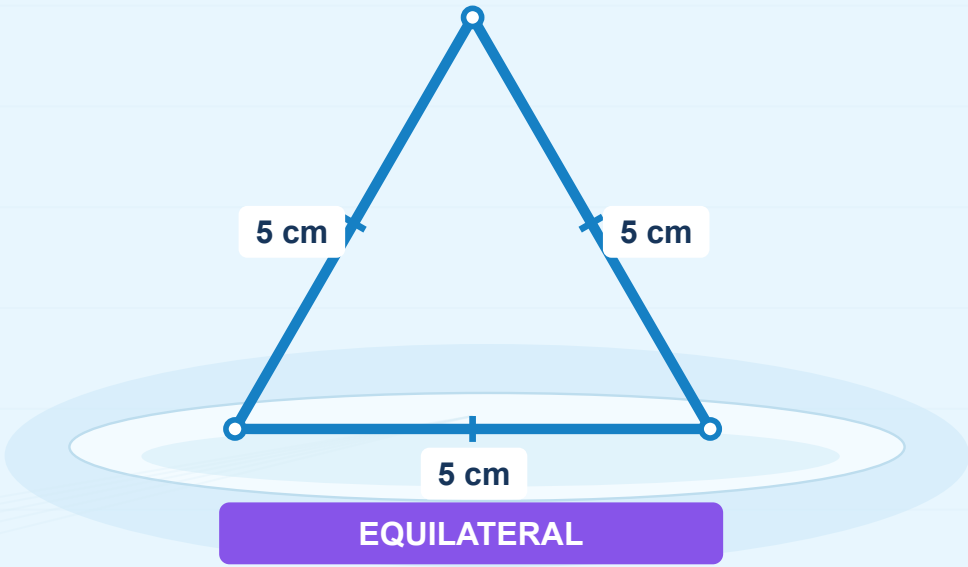
A triangle with two equal sides is called an isosceles triangle.

DRAG VERTEX



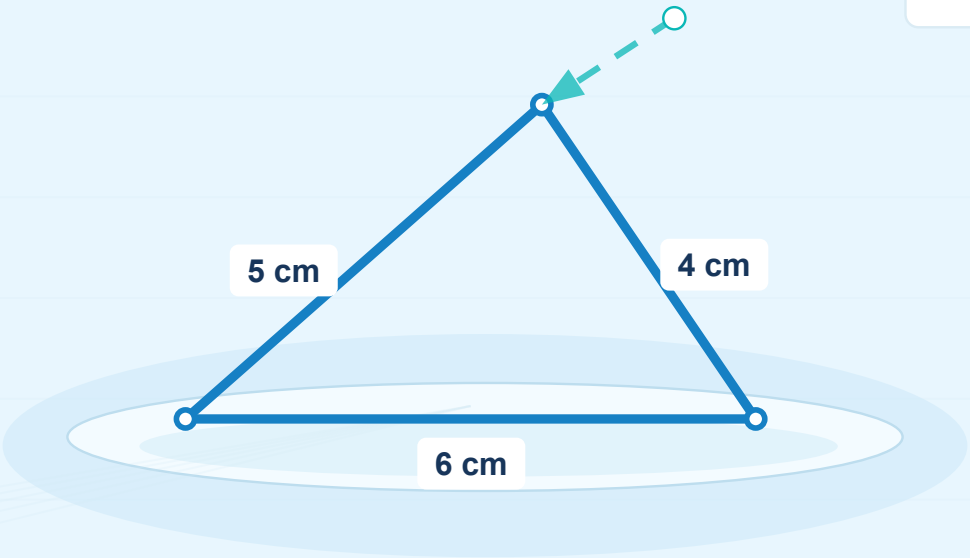
Can you make all three sides equal?

Drag a vertex until the three lengths match.



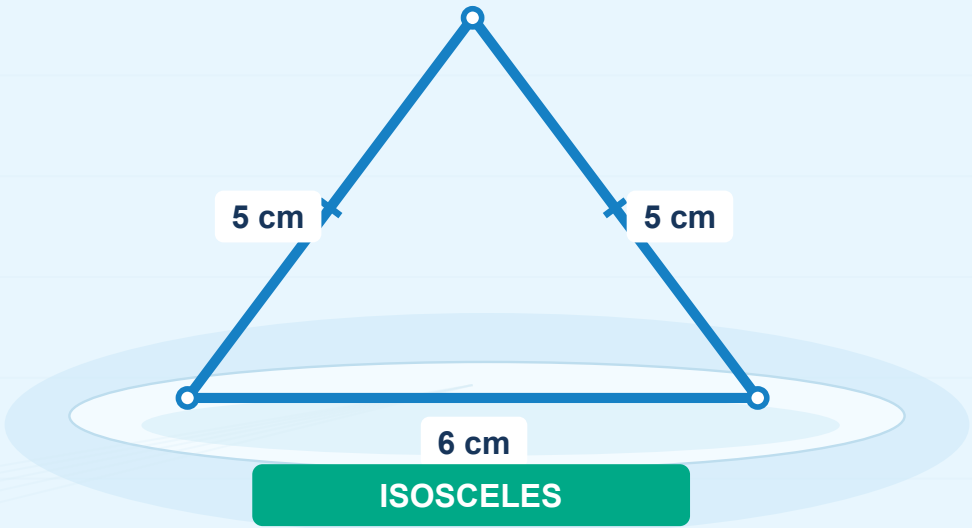
A triangle with all three sides equal is called an equilateral triangle.

DRAG VERTEX



Make this triangle isosceles.

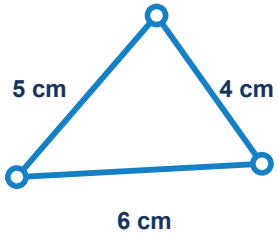
Move the vertices; no hints until an incorrect attempt.



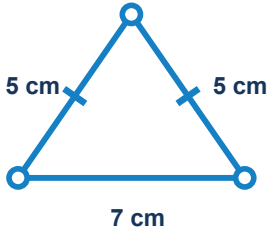
Great! This triangle is isosceles.

Sort the triangles.

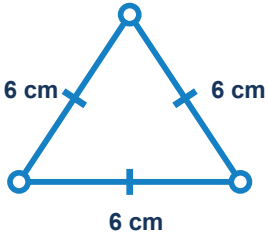
A



B



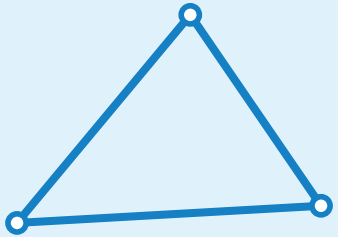
C



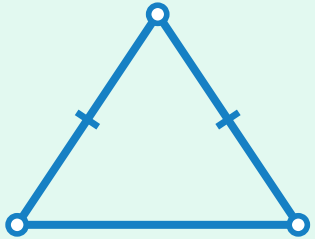
DRAG TO SORT

SCALENE	ISOSCELES	EQUILATERAL
DROP TRIANGLE HERE	DROP TRIANGLE HERE	DROP TRIANGLE HERE

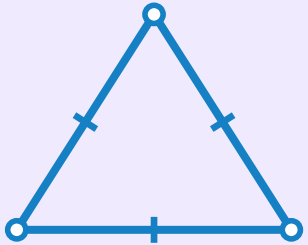
Great job! You sorted all the triangles.



SCALENE
5 / 4 / 6 cm



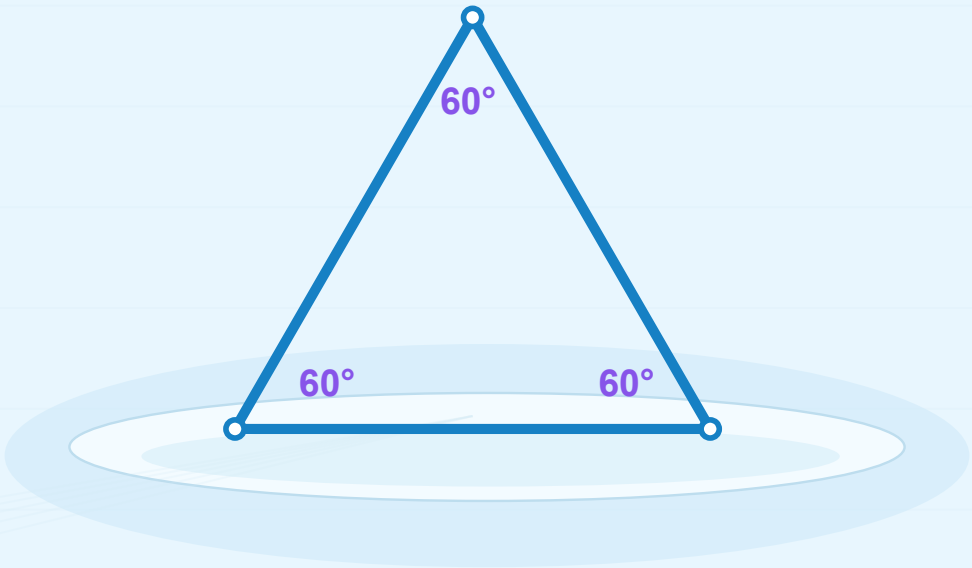
ISOSCELES
5 / 5 / 7 cm



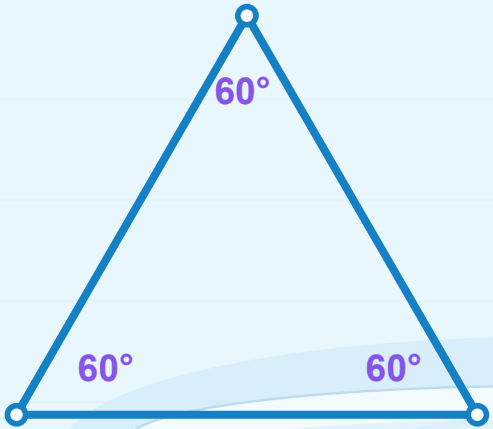
EQUILATERAL
6 / 6 / 6 cm

CONTINUE

ANGLE MEASURE

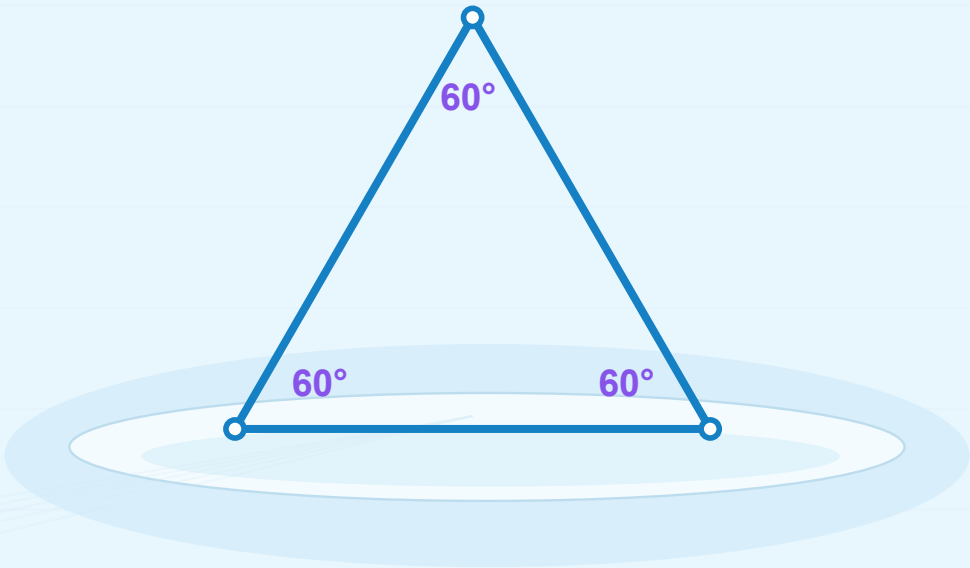


Now let's measure the angles.

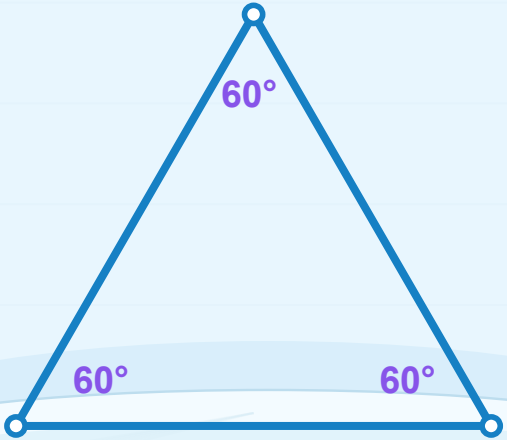


What do you notice about the angles?

- [x] All are acute angles
- [ ] One angle is right angle
- [ ] One angle is obtuse angle



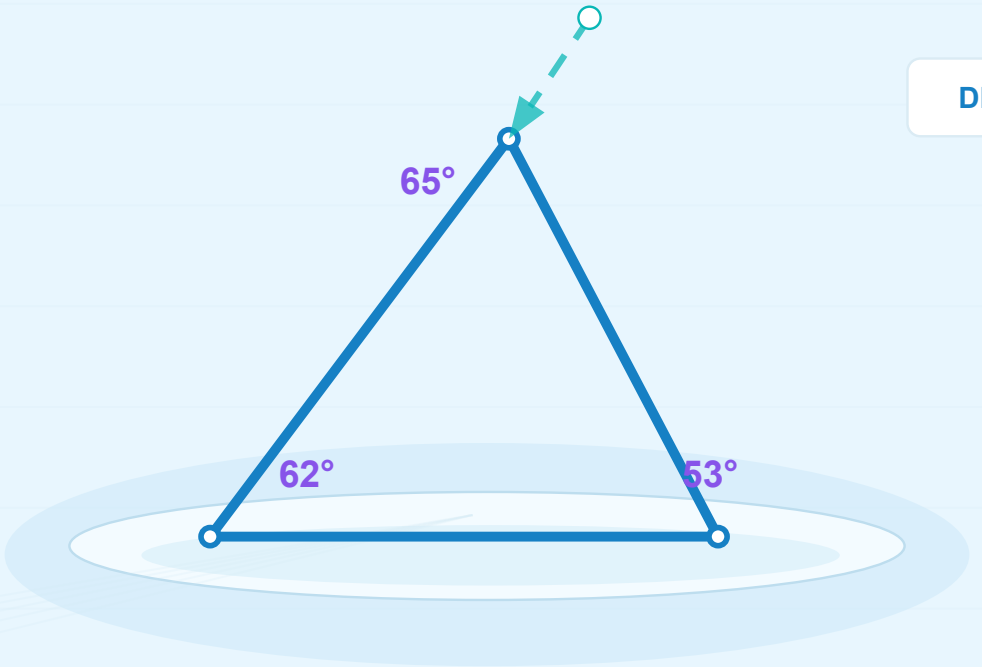
All three angles are less than 90°. So they are acute angles.



ACUTE-ANGLED

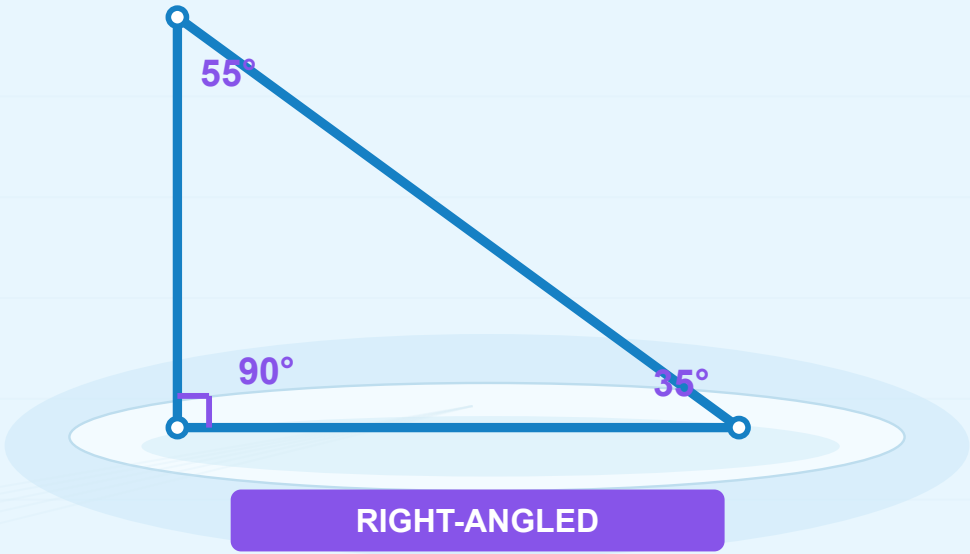
A triangle which has 3 acute angles is called an acute angled triangle.

DRAG VERTEX



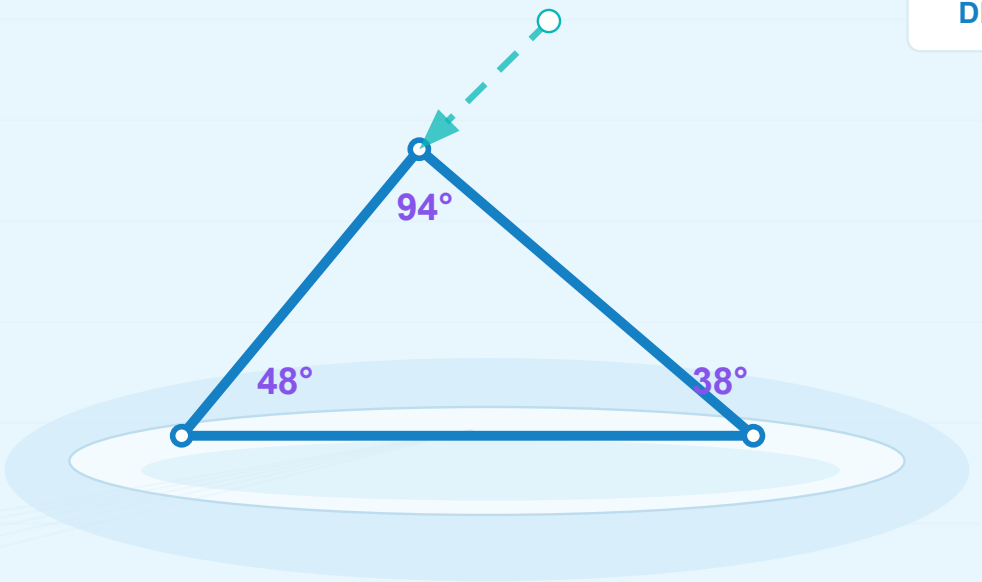
Can you make one angle exactly 90°?

Drag a vertex; angles change live.



A triangle with one angle equal to 90 is called a right angled triangle.

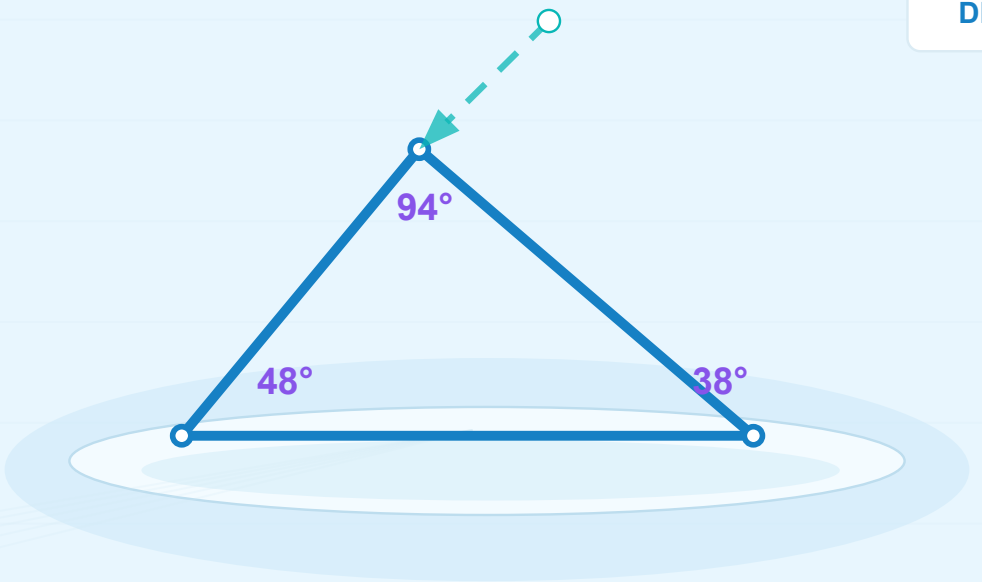
DRAG VERTEX



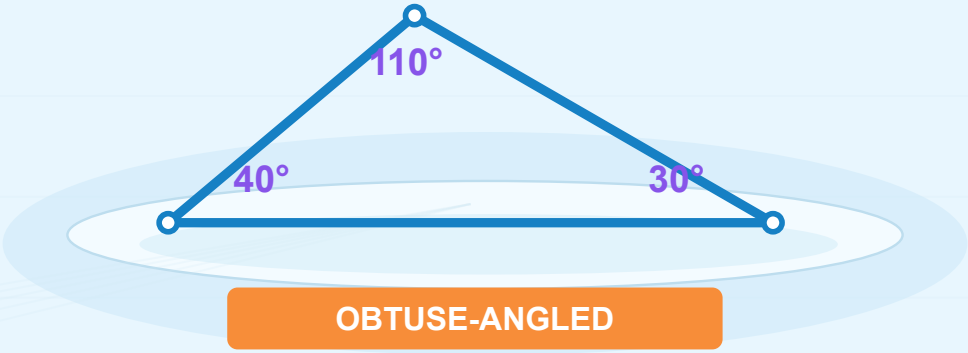
What if one of the angles was obtuse?

Drag any vertex to make any angle obtuse.

DRAG VERTEX



Drag any vertex to make any angle obtuse.

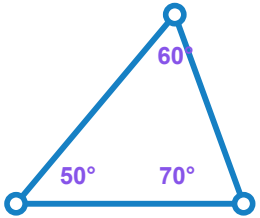


OBTUSE-ANGLED

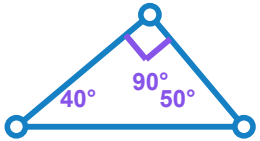
A triangle with one angle greater than 90° is called an obtuse-angled triangle.

Which of these are acute-angled triangles?

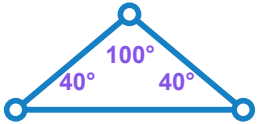
A



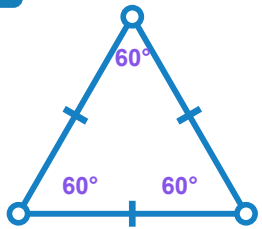
B



C



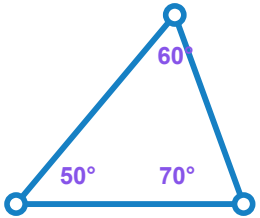
D



CHECK

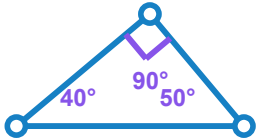
Correct! Both A and D have only acute angles.

A

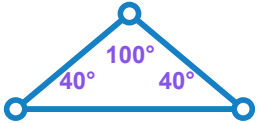


✓ SELECTED

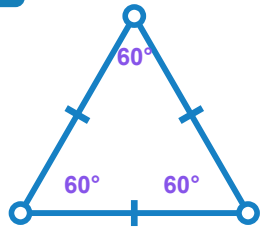
B



C



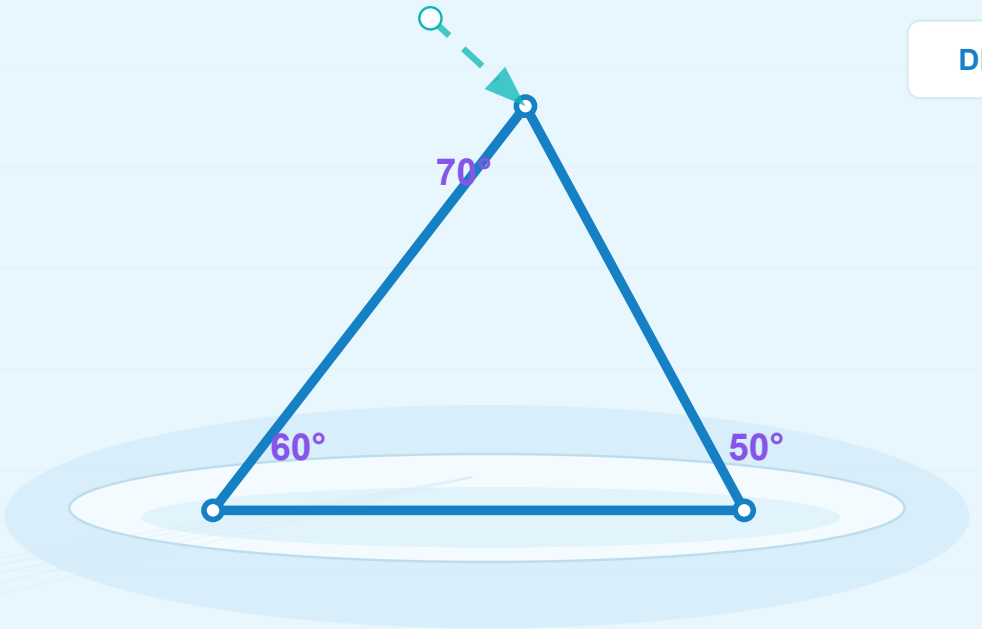
D



✓ SELECTED

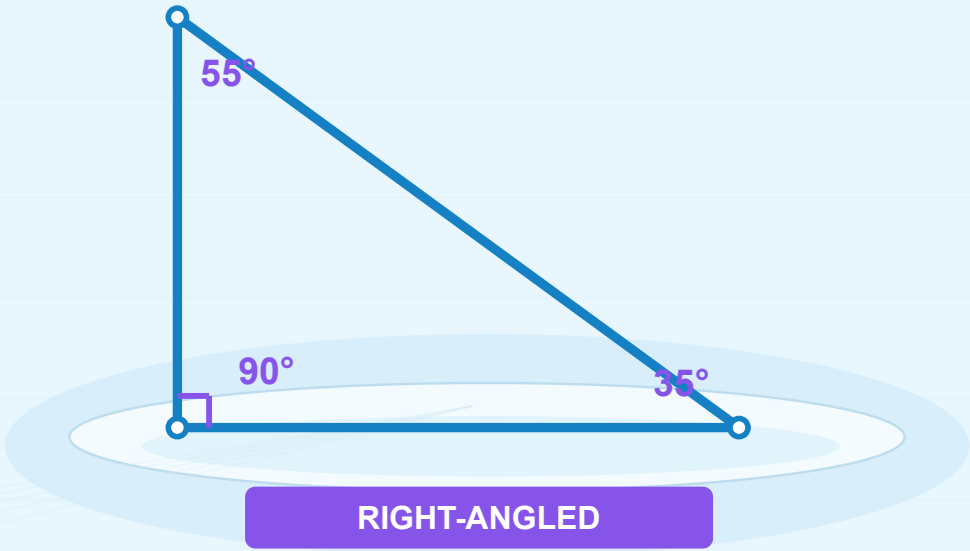
CONTINUE

DRAG VERTEX

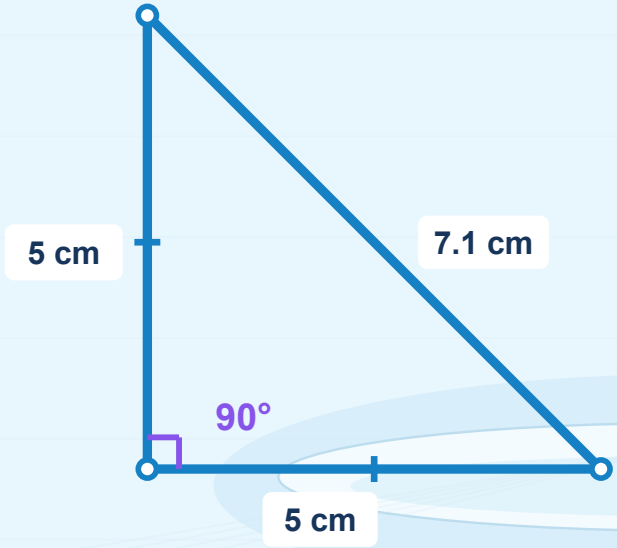


Make this triangle right-angled.

Drag any vertex to create one 90° angle.

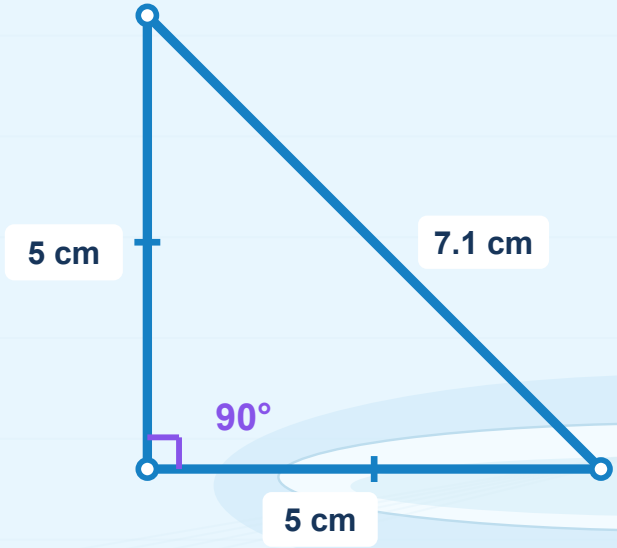


A triangle with one 90° angle is right-angled.



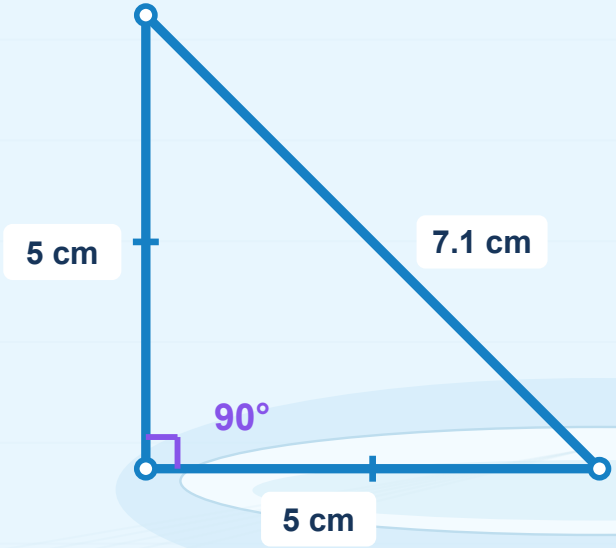
What type of triangle is this?

- ○ Scalene
- ● **Isosceles**
- ○ Equilateral



What type is this triangle by its angles?

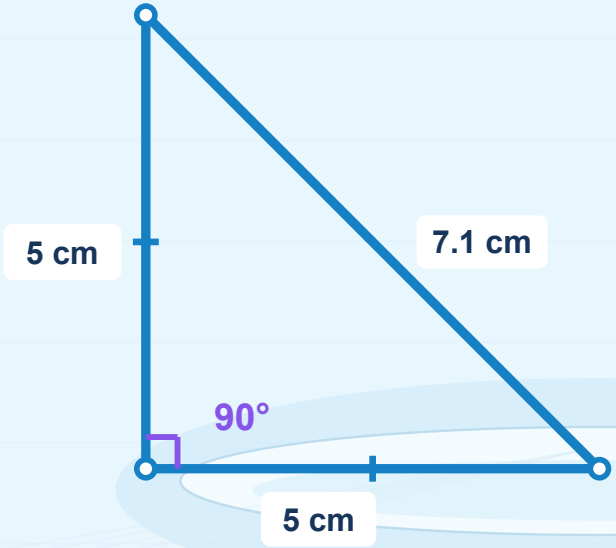
- ○ Acute-angled
- ● **Right-angled**
- ○ Obtuse-angled



ISOSCELES

RIGHT-ANGLED

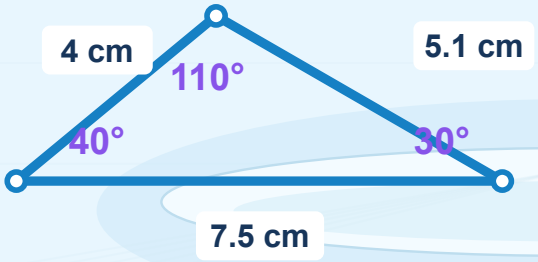
This triangle is both isosceles and right angled.



ISOSCELES

RIGHT-ANGLED

So a triangle can be classified by both its sides and its angles.



Classify by sides and angles.

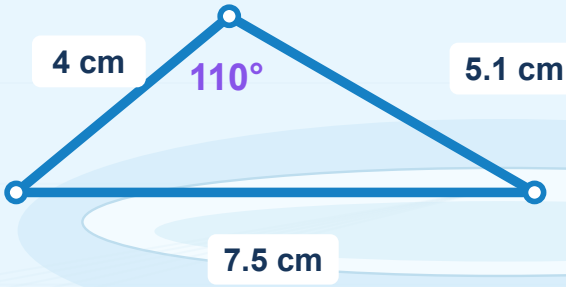
BY SIDES

- ◯ Scalene
- ◯ Isosceles
- ◯ Equilateral

BY ANGLES

Acute | Right | Obtuse

Classify the triangle.

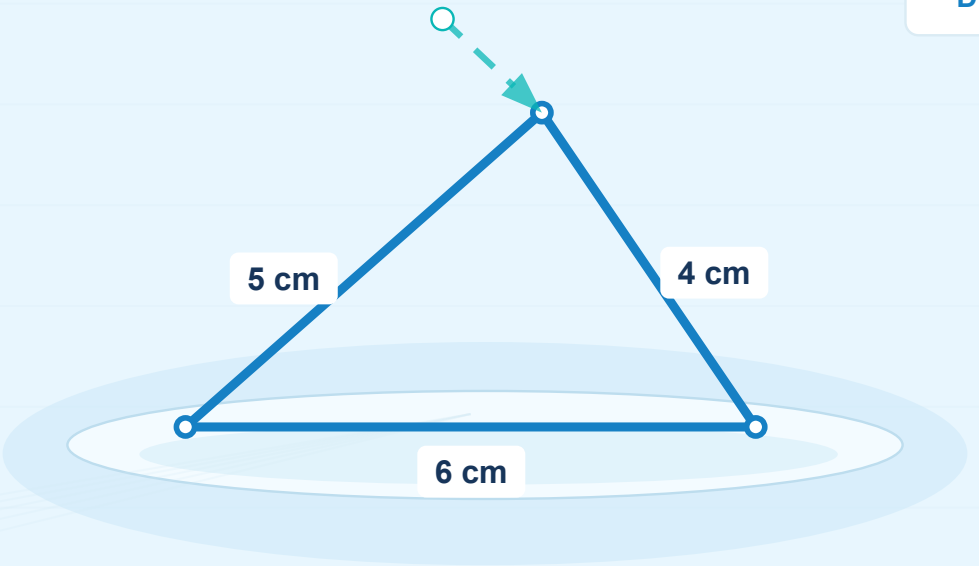


SCALENE

OBTUSE-ANGLED

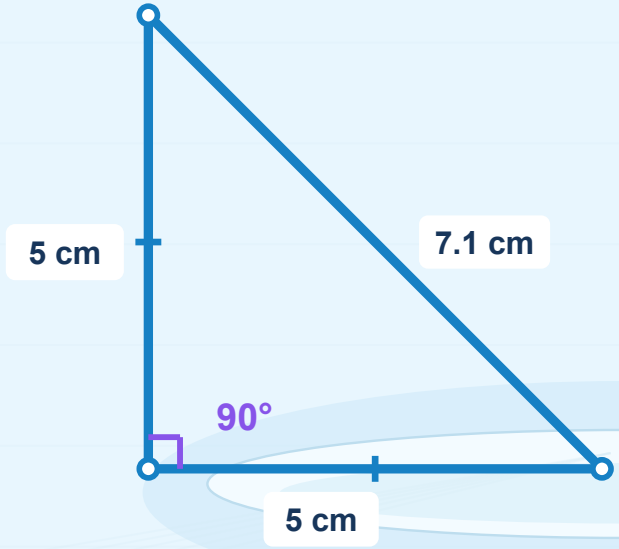
Correct! This triangle is scalene and obtuse-angled.

DRAG VERTEX



Make a right-angled isosceles triangle.

No hints: satisfy both conditions in one shape.



ISOSCELES

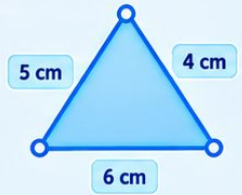
RIGHT-ANGLED

✓ GREAT JOB!

You can classify triangles by sides and angles!

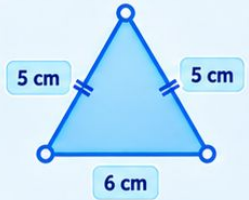
Let's summarise what we learned.

Triangles by sides



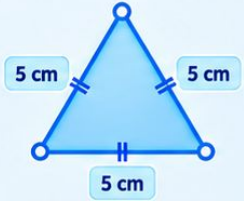
Scalene

All sides are different.



Isosceles

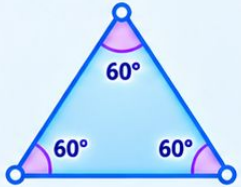
Two sides are equal.



Equilateral

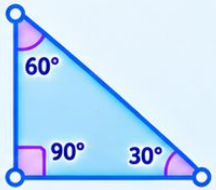
All three sides are equal.

Triangles by angles



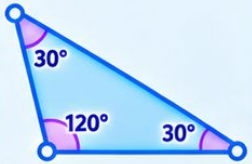
Acute-angled

All angles are less than 90°.



Right-angled

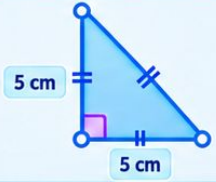
One angle is 90°.



Obtuse-angled

One angle is greater than 90°.

A triangle can be classified by both its sides and its angles.



Isosceles

Right-angled

You're ready! Let's protect the spaceship. →